

Enhancing Sign Language Learning Through Augmented Reality Feedback

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Abstract

Sign language education remains difficult to access for many learners, particularly those studying independently without access to expert feedback on gesture accuracy. Existing self learning resources, such as textbooks and videos, do not provide real-time correction, making it challenging for beginners to develop correct hand shapes and movements.

This project presents a browser-based augmented reality (AR) system for learning British Sign Language (BSL) that provides real-time visual and quantitative feedback on user performed gestures. The system integrates MediaPipe hand landmark tracking with a neural network classifier for static sign recognition and Dynamic Time Warping (DTW) for evaluating dynamic gestures. Users interact with the system through a webcam interface, receiving immediate feedback on hand pose, motion accuracy, and confidence scores.

The system was evaluated through a user study involving participants performing guided BSL gestures followed by a structured questionnaire. Results indicate that the AR-based approach improved users understanding of hand orientation, gesture clarity, and motivation to practise, with consistently high user ratings across these dimensions. Participants also reported increased engagement compared to traditional textbook based learning methods, highlighting the value of interactive and feedback-driven learning environments.

The findings demonstrate that lightweight, browser-based AR systems can effectively support independent sign language learning. This work contributes a practical framework for accessible BSL education and highlights the potential of combining real-time computer vision with interactive feedback to improve learning outcomes.

1 Introduction

Sign languages are essential for the Deaf and hard-of-hearing community, supporting accessible communication, social inclusion, and self advocacy. However, access to formal British Sign Language (BSL) learning remains limited due to shortages of qualified instructors and the cost and scheduling constraints associated with in-person training. As a result, many learners rely on self study resources such as diagrams or video tutorials, which often fail to provide real-time correction or support accurate movement learning [12]. Without immediate feedback on handshape, spatial placement, or temporal motion, beginners frequently develop errors that hinder progress and confidence.

Advances in computer vision (CV), machine learning (ML), and augmented reality (AR) provide new opportunities to address these challenges. AR encourages embodied learning by overlaying guidance directly within the user's physical environment, and prior research demonstrates improvements in retention and motivation when learners receive interactive and personalised support [12, 30]. At the same time, large scale datasets such as BSL-1K [2] and multimodal corpora like How2Sign [11] have enabled

advances in automatic sign recognition using spatial-temporal modelling techniques.

Despite this progress, accessible and deployable learning systems remain limited. Many state-of-the-art recognition approaches rely on computationally intensive models or controlled recording conditions, restricting their applicability in real-world educational settings [9]. Furthermore, existing tools often focus on static pose matching and do not adequately capture temporal dynamics, even though motion plays a critical role in distinguishing meaning in sign languages [13]. This highlights the need for lightweight, interactive systems that provide both spatial and temporal feedback in real time.

To address this gap, this project presents a browser based AR learning system designed to support independent BSL practice. The system integrates MediaPipe-based hand tracking with a neural network classifier for static gesture recognition, alongside a Dynamic Time Warping (DTW) approach for evaluating dynamic gestures. The application provides real-time feedback through visual overlays, confidence scores, and motion analysis, enabling users to iteratively refine their signing.

To evaluate the system, a user study was conducted in which participants completed guided sign language tasks using the application, followed by a structured questionnaire assessing usability, engagement, and perceived learning effectiveness. The evaluation combined quantitative Likert-scale responses with qualitative feedback to capture both measurable outcomes and user perceptions.

The results indicate that the AR-based approach improved the participants' understanding of hand orientation, gesture clarity, and motivation to practice. Users also reported higher engagement compared to traditional learning methods, highlighting the potential of interactive feedback systems to support independent skill acquisition. These findings suggest that lightweight browser-based AR technologies can provide an effective and accessible platform for sign language learning.

2 Problem Statement

Current BSL learning tools do not provide the real-time, error-corrective feedback necessary for effective independent acquisition of sign accuracy and fluency. Traditional resources lack personalisation, and existing automated systems often struggle with varied environments, generalising across different users, and recognising temporally-defined signs [9]. Moreover, many CV-based recognition solutions are not optimised for consumer-grade hardware or web deployment, restricting accessibility outside specialist settings.

There is a clear need for an interactive learning system that is able to:

- operate fully in-browser using efficient inference,
- guide beginners by detecting and explaining errors in static hand configurations, and
- eventually support dynamic gesture interpretation through temporal modeling.

This project addresses these needs by developing and evaluating a real-time AR-assisted BSL tutoring assistant that integrates lightweight ML-based hand-pose classification with interpretable instructional feedback. The intended contribution is a deployable prototype demonstrating that modern hand-tracking technologies and compact deep learning models can effectively support accessible sign language instruction for non-expert learners.

3 Background Survey

Sign languages are complex visual gestural communication systems with unique expressive styles, educational rules, and grammatical structures [27]. British Sign Language (BSL) relies on coordinated gestures, spatial positioning, orientation, movement, facial expressions, and body posture to convey linguistic meaning. These multichannel characteristics pose challenges for automated recognition due to occlusions, articulation uncertainty, and strong signer-dependent variation [1]. Even though the field of Sign Language Recognition (SLR) has advanced significantly, the majority of systems still find it difficult to function reliably in real-world learning environments where users lack controlled backgrounds or consistent gesture form.

3.1 Vision Based Sign Language Recognition

From early manual pipelines to modern deep learning algorithms, Vision based Sign Language Recognition (SLR) has advanced a long way. Traditional approaches often relied on skin colour segmentation, optical flow fields, and Histogram of Oriented Gradients (HOG) to extract motion cues from RGB video, followed by traditional machine learning algorithms such as Support Vector Machines (SVMs) or Hidden Markov Models (HMMs) [32]. Although these systems performed reasonably well in controlled settings, they performed poorly in real world variations, especially when exposed to barriers, crowded backgrounds, and various kinds of skin tones [4]. Furthermore, handcrafted pipelines did not prove durable to abnormal lighting circumstances and signer-specific syntax differences, which are inevitable in home learning contexts.

The shift toward deep learning has made it possible to extract hierarchical visual representations that are specific to gesture patterns. While 3D CNNs and two stream architectures directly incorporate spatial temporal motion dynamics from raw video frames, Convolutional Neural Networks (CNNs) have proven to perform well in static handshape identification [23, 28]. By incorporating long range temporal attention, transformer-based architectures have recently demonstrated state-of-the-art scalability in continuous SLR and sign-to-text translation tasks [8]. Large annotated datasets like BSL-1K [2], WLASL [18], and Phoenix2014T [13], that document variation in signing individuals, mobility patterns, and linguistic context, have emerged to support these deep models.

However, performance gains from end-to-end CNN/Transformer architectures come with substantial computational cost. Inference often depends on powerful GPUs or specialised accelerators due to high resolution inputs and large model sizes [28]. These constraints hinder deployability on mobile devices or in web based environments where learners expect real time responsiveness. Moreover, pixel based models remain sensitive to environmental noise such as camera angle shifts, sleeve color similarity, and partial occlusions, factors that are out of human control and can disrupt reliable sign recognition outside controlled settings [4].

The implementation of Lightweight skeletal models based on extracted 2D or 3D hand landmarks significantly reduces the computational load while improving generalisation to variabilities in skin tone and background conditions [20]. This encourages a shift away from raw video towards structured geometry suitable for efficient browser based machine learning, aligning closely with the goals of the current project.

3.2 Skeletal Modelling and Pose Feature Learning

Skeletal modelling has emerged as a dominant strategy in real-time sign recognition due to its efficiency and robustness compared to pixel based video analysis. Modern middleware such as MediaPipe Hands estimates 21 anatomically meaningful 2D/3D landmarks per hand at interactive frame rates [20], enabling structured representations of hand configuration independent of background texture, skin tone, or camera lighting. By operating on sparse geometry rather than dense images. Models avoid high-dimensional convolutional pipelines, resulting in reduced computational load and significantly improved flexibility to limited resource environments, such as smartphones and browsers.

Pose based descriptors further enable extraction of linguistically grounded features, such as joint rotation angles, inter-finger distances, palm orientation, and hierarchical kinematic constraints [31]. These characteristics are particularly valuable in sign languages where language correctness is defined by hand-shape, location, and movement relative to the body [7]. Angle based features capture internal movement (e.g., finger bending) that raw landmark coordinates fail to distinguish, improving recognition of minimal pairs and duplicate forms. Research suggests such features align more closely with perceived clarity for learners and sign-language instructors [27].

Enhanced generalisation across users and perspectives is a further advantage of skeletal modelling. Normalisation techniques including scale alignment, reference point anchoring (e.g., wrist as anatomical root), and mirroring compensation for left/right hand dominance can reduce noise caused by partial interruptions or lateral camera angles. These steps enable interpretable, domain specific inference without the need for extensive data augmentation.

From an engineering perspective, compact posture attributes provide high compatibility with web-based inference engines like WebAssembly and ONNX Runtime Web, which need predictable execution routes and minimal memory consumption for real-time responsiveness [26]. Furthermore, unlike deep CNN activations, joint level deviations can be visualised and communicated as instructional guidance, which is a crucial requirement for personalised tutoring systems [5]. The skeleton structure also directly assists with the explainability and feedback creation.

This project's emphasis on accessible, browser-based BSL learning is greatly supported by skeletal modeling's combination of linguistic relevance, computational efficiency, and interpretability qualities, which make it an ideal basis for sign language tutoring.

3.3 Temporal Dynamics and Motion Interpretation

Temporal modelling remains essential in sign language recognition because many signs are distinguished not only by their static handshapes but by the motion trajectory, speed, repetition, and

start–end spatial locations [13, 33]. Signs that appear visually similar in a single frame can have entirely different meanings once temporal evolution is considered, such as the difference between polite BSL gestures like “*thank you*” and “*please*”, where motion direction and speech length act as key differences [7]. Failure to integrate movement cues can therefore, lead to uncertainty and lower the overall learner feedback quality.

Modern computer vision models address temporal variability using recurrent based neural architectures. Long Short Term Memory (LSTM) networks capture sequential dependencies and have demonstrated strong performance for continuous SLR tasks [8]. More recently, Transformer architectures enable richer temporal context and global modelling of gesture sequences, achieving state-of-the-art recognition in large video corporations [8]. However, these approaches are computationally intensive and require large datasets to generalise effectively across different learners and signing styles.

Traditional positional techniques, such as Dynamic Time Warping (DTW), offer an alternative that is lightweight, data efficient, and particularly well suited to user driven educational feedback [24]. DTW allows temporal stretching and compression to account for natural signing variability while still preserving the linguistic structure of gestures. Because beginners often perform signs, either more slowly or with inconsistent timing compared to more fluent signers, DTW provides meaningful similarity scoring without imposing strict timing constraints. As such, DTW remains highly relevant for deployment-constrained environments and for real time tutoring systems, which must provide interpretable feedback. This project utilises DTW as a useful starting point for modelling motion based BSL gestures, allowing an AR learning interface to automatically assess temporal consistency.

3.4 Datasets for Sign Recognition and Learning

The advancement of automatic recognition has been greatly aided by the availability of large sign language datasets. The small size and difficult management of early datasets limited their usefulness to real-world generalisation. Scalable, diverse collections that capture natural signature variation have become the focus of recent academic projects. For instance, research into broad vocabulary classification was made possible by BSL-1K, which introduced more than 1,000 British Sign Language (BSL) sign classes derived from broadcast video [2]. For sign-to-speech translation and contextual gesture understanding, How2Sign offers a complete multimodal dataset that connects RGB video, depth streams, and motion capture [11].

Despite having advanced recognition systems, these datasets aren’t specifically made for educational purposes. They usually include fluent signers rather than inexperienced learners, which limits the model’s capacity to accept linguistic mistakes or uncertainty typical of early skill acquisition. Additionally, many datasets are less suitable for feedback based correction or browser embedded landmark inference because they lack deep skeletal or linguistic annotations like handshape category and signer viewpoint. Alignment metadata, such as start and finish frames of important movement phases, is frequently lacking or inconsistent among datasets, even in cases when temporal dynamics are provided.

Another limitation relates to deployment and accessibility. Large multimodal datasets are difficult to implement in lightweight, WebAssembly-based educational systems since they usually require high bandwidth processing and storage [26]. Additionally, licensing limitations on broadcast video, like those seen in BSL-1K, can make it more difficult to integrate it into public facing applications, particularly those that facilitate large scale digital learning.

These gaps highlight the need for datasets specifically tailored for instructional goals rather than recognition accuracy alone. Specifically, future datasets should prioritise:

- (1) inclusion of slow, learner appropriate signing styles,
- (2) educationally grounded accurate labels,
- (3) rich annotations for landmarks that are compatible with browser inference and,
- (4) varied camera angles to reflect home learning conditions

Addressing these requirements would strengthen the bridge between academic SLR research and deployable real-time tutoring systems for new BSL learners.

3.5 Sign Language Learning Technologies

E-learning platforms for sign language often rely on pre-recorded videos or static diagrams, which provide no error correction or visual feedback [15]. However, learning research highlights the importance of embodied, feedback-driven rehearsal for motor-skill development [16]. Augmented reality systems can therefore provide meaningful learning advantages by anchoring feedback in a user’s own visual context [6].

Recent AR-assisted prototypes demonstrate improved motivation and gesture refinement through real-time correction [10, 30]. Yet, many rely on mobile GPUs or marker based tracking, diminishing accessibility. VR-based signing tutors [5] demonstrate immersive interaction but require specialised devices that limit adoption.

A persistent gap exists between high-performing recognition research and deployable, everyday sign language education tools. The system proposed in this research addresses this disconnect by:

- leveraging lightweight landmark-based feature extraction,
- integrating interpretable angle-based feedback aligned with sign phonology,
- utilising ONNX Runtime Web for browser-based, cross-device inference,
- and incorporating temporal scoring extensions for dynamic signing.

This positioning differentiates the proposed approach as both technologically feasible and educationally grounded, strongly aligned with the accessibility goals of inclusive digital learning.

3.6 Gaps and Opportunities

Sign Language Recognition (SLR) research has produced impressive results in controlled laboratory settings, however the conversion of these developments into real-world educational materials continues to be limited and scattered [4, 15]. Current systems frequently prioritise benchmark evaluation above learner experience, which creates a gap between the practical requirements of new learners pursuing self-guided study and the latest developments in recognition research.

There still exist multiple substantial gaps in the research and deployed technologies:

- (1) **Lack of accessible learning platforms:** Deployment in browser-based or low-resource contexts is hindered by the majority of existing systems' reliance on native mobile applications or strong GPUs [26]. Many students who simply have access to standard consumer gadgets are forced excluded from this.
- (2) **Limited meaningful real-time feedback:** Many prototypes, recognition output is limited to binary correctness judgements, with few details about *why* an error happened or *how* to improve [14]. According to educational studies, inexperienced students need educationally aligned input for them to develop proper hand shapes and motion patterns.
- (3) **Insufficient support for dynamic gesture evaluation:** Due to their simpler modelling, isolated static signs continue to have a large bias; nevertheless, a significant number of meaningful indications depend on temporal motion, directional movement, or volume variation [13]. Systems that ignore temporal dynamics are unable to scale to cover the entirety of the BSL curriculum.
- (4) **Poor adaptability under real-world changes:** When models are used outside of clean datasets, variations in signer anatomy, backdrop chaos, lighting, occlusion, and camera positioning reduce performance [2]. Household learning environments, where conditions cannot be strictly regulated, are rarely specifically addressed by systems.

Together, these shortcomings highlight a significant area for innovation: the creation of a fully browser-based BSL teaching system that can provide inexperienced students with thorough, responsive, and academically relevant feedback. The suggested approach attempts to close the gap between technical recognition performance and successful language learning by utilising lightweight skeletal elements for fast inference and enhancing them with educational signals and temporal comparison techniques. This strategy promotes equal access to sign language learning and might significantly advance the Deaf community's digital inclusion.

4 Proposed Approach

The primary objective of this project is to provide an engaging, augmented reality-based British Sign Language (BSL) learning environment. Learners actively practise signs in front of their webcam instead of simply taking in educational content, and the technology delivers them immediate real-time feedback on hand pose accuracy, signing posture, and motion correctness. The AR interface encourages kinaesthetic learning through physical contact by superimposing visual guidance cues directly into the learner's field of view [6]. In a publicly accessible, browser-based setting, the system aims to mimic important elements of human coaching, including encouragement, organised growth, and constructive criticism. The end-to-end system architecture is illustrated in Figure 5, while the complete learner interaction lifecycle is summarised in the high-level AR interaction flow shown in Figure 1.

To realise this vision, the proposed implementation integrates lightweight machine learning models with real-time computer

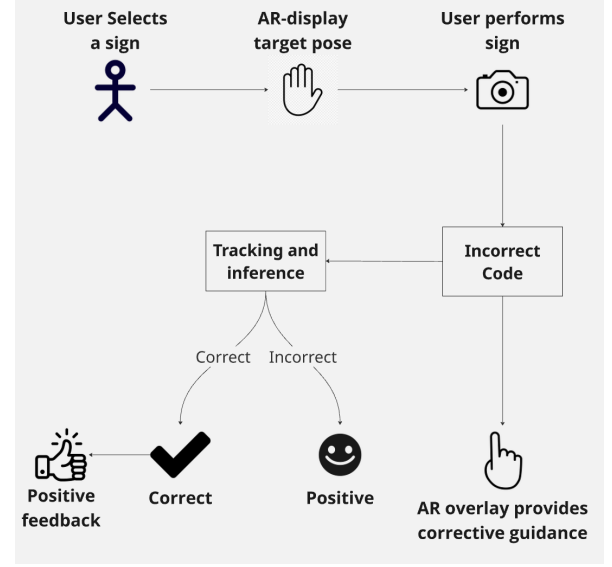


Figure 1: AR Interaction Flow Diagram



Figure 2: Data Processing & Feature Representation Diagram

vision in an interactive learning application. The following illustrations highlight the essential elements within the proposed methodology.

4.1 Static Gesture Recognition via Hand Landmark Classification

For alphabetic and fixed pose signs, the system uses MediaPipe Hands to extract 21 hand landmarks in real time [20]. These 3D coordinates are flattened into feature vectors; 63 dimensions for single hand signs and 126 dimensions for two hand signs, and classified using a compact fully connected neural network deployed through ONNX Runtime Web [26]. Lightweight inference ensures broad device compatibility without requiring GPUs or native software installation. Figure 2 summarises the data processing steps, showing how raw MediaPipe landmarks are normalised, transformed, and fed into the neural classifier.

The system returns a sequence score indicating similarity between the user's posture and the target BSL sign, enabling clear and interpretable feedback. An example of the live practice interface and sequence scoring is shown in Figure 4

4.2 Temporal Gesture Assessment via Dynamic Time Warping

Following similar techniques employed in lightweight recognition prototypes, the system utilises Dynamic Time Warping (DTW) to compare user motion sequences to generate patterns for dynamic signs incorporating motion trajectories (such as signalling gestures) [13, 34]. Sequences are evaluated using a

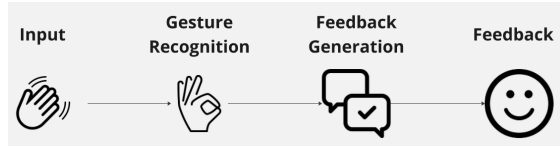


Figure 3: Feedback System Diagram

normalised matching score after being resampled for temporal alignment.

Explanation:

- DTW enables a range of frequencies
- It performs well with small datasets
- Neural classifiers for motion encoded gestures are linked

4.3 Real Time Feedback Mechanism

A key component of learning sign language is receiving immediate corrective feedback. The proposed methodology assesses:

- Colour coded skeleton alignment indicators for pose correctness
- Text prompts for hand positioning (e.g. “raise palm”, “move closer to camera”)
- Sequence scores and predicted sign labels for immediate user awareness
- Motion progress indicators during dynamic sign attempts

This design supports self correction, more frequent practice, and stronger engagement of the user compared to traditional resources [14]. The flow of structured feedback, from detection to user facing guidance, is illustrated in Figure 3. The full practice workflow, from landmark input to model output, is shown in the training and deployment pipeline displayed in Figure 6.

4.4 Deployment and Accessibility Strategy

The application runs fully in the browser using WebAssembly acceleration, enabling:

- Zero install access across laptops, tablets, and mobile devices
- Privacy preservation by keeping all inference on device
- Instant scalability, no server compute or login required

This aligns with the project’s commitment to inclusivity and universal access for diverse user groups.

4.5 Feasibility and Potential Risks

The significant integration of lightweight neural inference technologies and reliable 3D hand tracking pipelines makes the suggested technique physically possible. In real time gesture recognition systems, landmark based classification has proven to be a dependable method that maintains high accuracy while requiring little processing power. Because of this, browser based execution with ONNX Runtime Web is both practical and scalable on a variety of user devices.

It is also possible to score gestures dynamically using DTW, especially in applications where datasets can be limited. Learners can execute gestures with natural variation thanks to DTW’s ability to deal with temporal variations like different signing speeds. Because of its integration, a wider range of BSL signs that rely on motion rather than just static handshape can be supported by the system.

But there are certain risks to be aware of. First, visual occlusions, low lighting, and hand dominance (left vs. right performance) can still affect system reliability and result in incorrect classification or low sequence scores. Second, the size of the current dataset restricts generalisation across various learners, signing styles, and demographic variations, a problem that is well known in the field of sign language recognition research [11]. Third, even though WebAssembly allows for high-performance on-device processing, low power devices may still face latency or lower frame rates, which could negatively impact user experience.

Progressive dataset growth (e.g., adding BSL-1K for better generalisation), verification procedures to accommodate left handed users, and dynamic performance adjustment based on real time browser capability detection can all help reduce these risks. The system’s long-term scalability will be enhanced by ongoing iteration and community-driven data collection.

5 Evaluation

This section presents the methodology and results used to evaluate the effectiveness of the proposed AR-based sign language learning system. The evaluation focuses on both the technical performance of the system and its educational value for beginner learners.

5.1 Study Aims and Research Questions

The aim of this evaluation is to determine whether an augmented reality (AR) based learning system can effectively support beginner learners of British Sign Language (BSL), particularly in comparison to traditional self learning methods.

To guide the evaluation, the following research questions were defined:

- **RQ1:** Does the AR system improve the understanding of user orientation and gesture structure?
- **RQ2:** Does real-time feedback improve users perceived ability to perform sign language gestures accurately?
- **RQ3:** Does the AR learning approach increase user engagement and motivation compared to traditional learning methods?
- **RQ4:** Do users perceive the system as an effective and accessible tool for independent sign language learning?

In addition to these user-focused outcomes, the evaluation also considers the reliability of the gesture recognition system, as accurate feedback is essential to ensure that users learn correct sign representations rather than reinforcing incorrect movements.

5.2 Study Design

To investigate these questions, a user centered evaluation methodology was adopted. The study used a mixed methods approach, combining:

- **Quantitative data:** responses of the Likert scale questionnaire measuring user perceptions
- **Qualitative data:** Open ended feedback describing user experiences

This approach allows both measurable trends and deeper insights into user experience to be captured, providing a more comprehensive evaluation of the system.

User studies are widely used in Human Computer Interaction (HCI) research to evaluate interactive systems, as they provide information on how real users interact with new technologies in

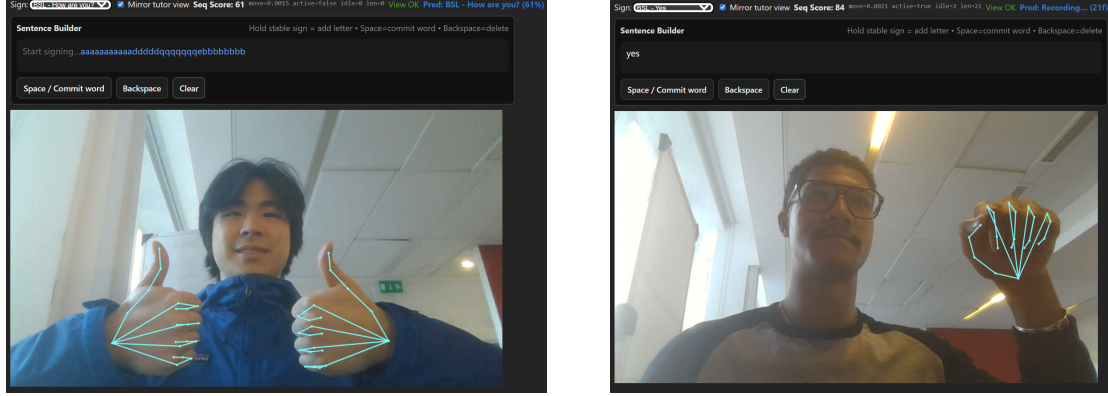


Figure 4: BSL practice interface showing MediaPipe hand detection and sequence scoring during sign execution.

practice [17]. Previous work in sign language learning technologies highlights the importance of user perception, particularly in areas such as feedback clarity, usability, and perceived learning effectiveness [10].

Although engagement and motivation are important factors in learning, this study also emphasizes the importance of learning accuracy, as incorrect acquisition of gestures can negatively impact long-term skill development. Therefore, both usability and perceived learning correctness are considered in the evaluation.

5.3 Study Procedure

The evaluation consisted of two main phases: an interaction phase and a feedback phase.

Interaction Phase

Participants were introduced to the AR learning system and asked to practise a series of BSL gestures using the application. During this phase, the system provided:

- Real-time hand tracking using MediaPipe
- Gesture recognition via a neural network classifier
- Immediate feedback through confidence scores and visual overlays

Participants were encouraged to repeat gestures multiple times in order to explore the system’s feedback and improve their performance.

Feedback Phase

After completing the interaction tasks, participants were asked to complete a structured questionnaire evaluating their experience. The questionnaire assessed:

- Gesture clarity and understanding
- Feedback effectiveness
- Cognitive effort required
- Engagement and motivation
- Comparison with traditional learning methods
- Perceived accessibility of the system

These measures are commonly used in evaluations of educational technologies, as they provide insight into how effectively a system supports learning processes [6].

This evaluation provides an initial assessment of the feasibility and educational potential of AR-assisted sign language learning. The following sections present the experimental setup, measurement approach, and results obtained from the user study.

5.4 Experimental Setup

A total of 14 participants participated in the evaluation. Participants were recruited from the University of Glasgow and had no formal training in British Sign Language (BSL), ensuring that the study focused on beginner level learning scenarios. This was important, as the system is specifically designed to support independent learners who may not have access to expert instruction or prior experience.

The evaluation was designed to simulate a realistic self-learning environment, where users interact with the system without direct supervision. Participants were first introduced to the system and given a brief explanation of the study procedure. They were then asked to complete a set of guided interaction tasks using the AR based learning interface.

The gesture set used in the study included a small number of representative BSL signs, selected to cover both static gestures (e.g., alphabet signs such as “A”) and dynamic gestures (e.g., greetings such as “Hello”). These were chosen to evaluate the system’s ability to support both handshape recognition and motion-based gesture understanding. The selection was intentionally limited to ensure that participants could complete the tasks within a short session while still experiencing the core functionality of the system.

During the interaction phase, participants were first shown example gestures through the application interface. These demonstrations were presented visually via the system, allowing users to observe the intended hand shape and movement before attempting to replicate them. Participants were then instructed to perform each gesture using the webcam based interface.

For each gesture, participants were encouraged to repeat their attempts multiple times in order to improve their performance based on the feedback provided. No strict performance threshold (e.g., a required confidence score) was enforced; instead, the study aimed to observe how users naturally responded to the system’s feedback and whether they were able to iteratively refine their gestures. This approach reflects real-world self-guided learning, where users rely on feedback rather than formal assessment criteria.

The system provided real-time feedback through three main mechanisms:

- **Gesture recognition output**, indicating the predicted sign label
- **Confidence scores**, representing the system’s certainty

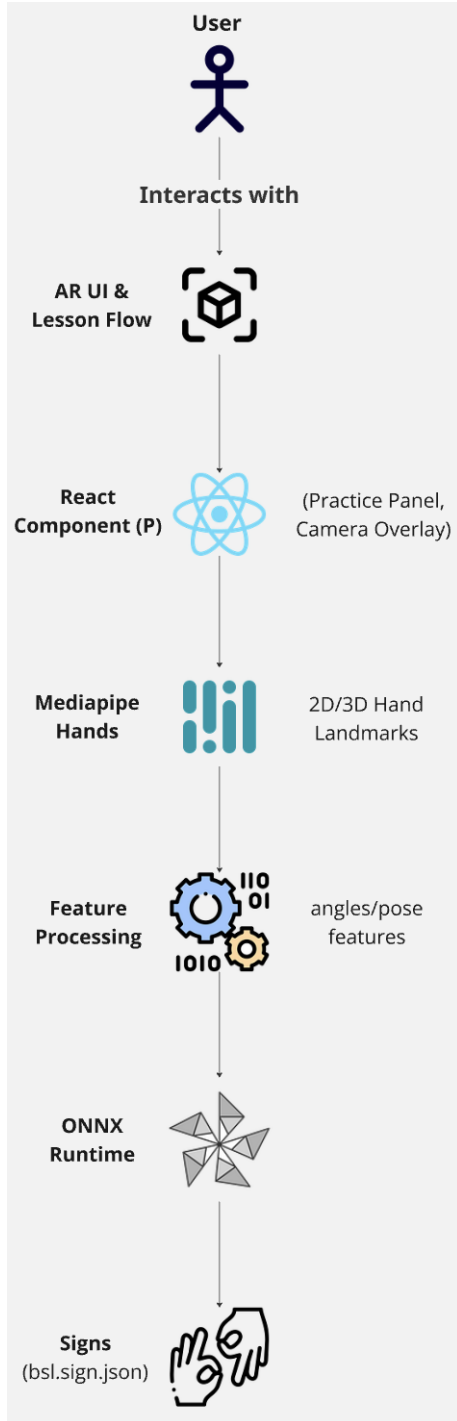


Figure 5: Overall System Architecture Diagram

- **Visual overlays**, showing hand landmarks and tracking information

Participants were not given detailed instructions on how to interpret these feedback signals. Instead, they were allowed to form their own understanding of how the system responded to their gestures. This design choice was intentional, as it allowed the evaluation to capture how intuitive and self-explanatory the feedback mechanisms were for first-time users.

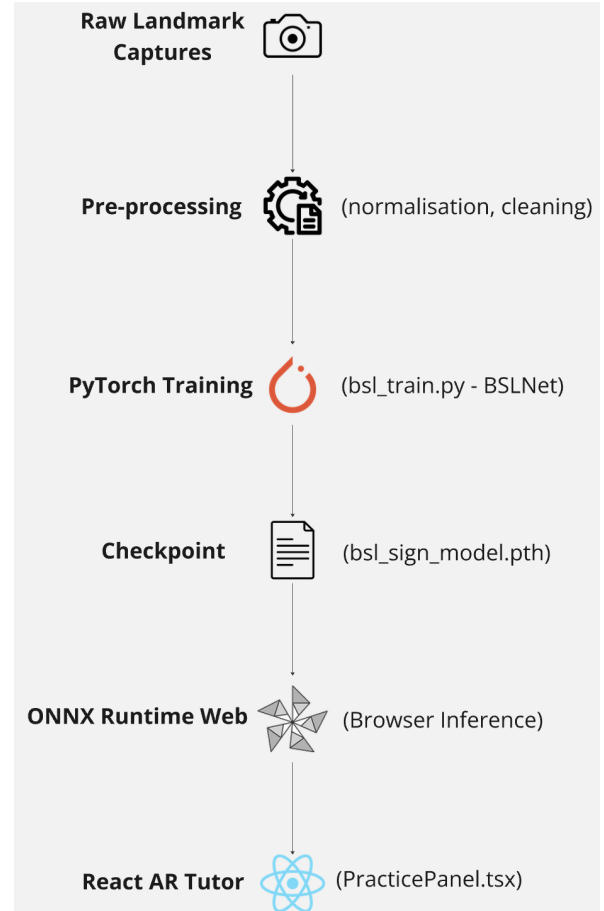


Figure 6: Training and deployment workflow for the BSL sign recognition model

Participants were also encouraged to explore the interface features during the interaction phase, including gesture visualisation and tracking overlays. This exploration was not treated as a separate task but rather as part of the natural interaction process, allowing participants to engage with the system in a flexible and user-driven manner.

Overall, this experimental setup was designed to balance structure and realism, ensuring that participants completed consistent tasks while still reflecting how the system would be used in independent learning contexts.

5.5 Experimental Measurements

To evaluate the effectiveness of the proposed system, a combination of quantitative and qualitative measures was used. The evaluation focused primarily on user perceptions of learning support, usability, and engagement, as these are critical factors in determining whether an educational technology is effective in practice. While objective performance metrics (e.g., recognition accuracy or gesture correctness) are important for system validation, prior research in Human–Computer Interaction (HCI) highlights that user experience and perceived learning support are equally important when evaluating interactive learning systems [10, 17].

Following the interaction phase, participants completed an online questionnaire designed to capture their experience with the system. The questionnaire consisted of Likert scale questions

(rated from 1 = Strongly Disagree to 5 = Strongly Agree) and optional open-ended responses [19]. This approach enabled the collection of both structured quantitative data and more detailed qualitative insights.

The questionnaire items were designed specifically for this study but were informed by commonly used evaluation dimensions in educational technology research, including usability, engagement, feedback effectiveness, and perceived learning support. These dimensions are widely used when evaluating interactive and immersive learning systems, particularly in AR-based educational environments [6]. The questions were constructed to align closely with the core features of the system, ensuring that each component of the application (e.g., gesture visualisation, real-time feedback, and interaction design) was explicitly evaluated.

The Likert scale questionnaire included the following statements:

- The AR interface helped me clearly understand the orientation of the hand signs
- Viewing the signs from multiple angles improved my understanding
- I was able to distinguish subtle finger positioning details
- The precision tracker on the app helped improve my sign language practice
- I had to concentrate very hard to understand the signs
- The AR method kept me more engaged than textbook learning
- I was motivated to continue practising sign language
- The system could make sign language learning more accessible
- I would prefer learning sign language using AR over traditional textbook methods

It should be noted that participants were not required to complete a controlled textbook based learning task as part of the study. Instead, the comparison to traditional methods was based on participants prior experience with learning from static materials (e.g., diagrams or videos). This approach allowed the study to capture perceived differences in engagement and effectiveness, although it introduces some subjectivity into the comparison.

These measures were designed to evaluate several key aspects of the learning experience:

- **Gesture clarity**, including understanding of hand orientation and finger positioning
- **Feedback effectiveness**, particularly the usefulness of the precision tracking system
- **Cognitive effort**, reflecting the mental demand required to interpret and perform gestures
- **Engagement and motivation**, which are critical for sustained learning
- **Perceived usefulness** and accessibility, especially for independent learners
- **Preference compared to traditional learning methods**

In addition to the quantitative questionnaire data, participants were invited to provide open-ended feedback describing their experience with the system. These qualitative responses were analysed using a thematic analysis approach, where recurring patterns and common themes were identified across participant responses. This method allowed for a deeper understanding of user perceptions, highlighting both strengths of the system (e.g.,

interactivity and feedback) and areas for improvement (e.g., clarity of instructions or recognition limitations).

Although the evaluation primarily focused on subjective user experience measures, limited system-level observations were also considered during analysis, such as how participants responded to feedback and whether they attempted to correct their gestures iteratively. These observations provided additional context to support the interpretation of the questionnaire results.

Overall, this mixed-methods approach enabled a comprehensive evaluation of the system by combining quantitative ratings with qualitative insights, allowing both measurable trends and user perspectives to be captured.

5.6 Questionnaire Results

The questionnaire responses were analysed using descriptive statistical methods, where mean scores were calculated for each Likert scale item (1 = Strongly Disagree, 5 = Strongly Agree). These values were used to assess overall participant perceptions of the AR learning system across key evaluation dimensions.

A summary of the results is presented in Table 1, which reports the mean score for each evaluation statement.

Overall, the responses indicate a strongly positive perception of the prototype, particularly in relation to engagement, gesture clarity, and motivation to practise.

As shown in Table 1, participants reported that the AR interface supported their understanding of hand orientation ($M = 4.33$) and benefited from viewing gestures from multiple angles ($M = 4.33$). These results suggest that the spatial visualisation capabilities of the system were effective in helping users interpret sign structure.

Similarly, the ability to recognise subtle finger positioning details also received a high mean score ($M = 4.33$), indicating that the hand tracking and skeletal representation provided sufficiently clear visual feedback for most participants.

The precision tracking component received a slightly lower, but still positive, mean score ($M = 4.11$), as shown in Table 1. This suggests that while the feedback system was generally helpful, there may be some variability in how users perceived its effectiveness when refining their gestures.

Engagement and motivation were among the highest-rated aspects of the system. Participants reported high agreement with statements relating to engagement during practice ($M = 4.33$) and motivation to continue practising sign language ($M = 4.44$), the highest score observed. These findings indicate that the interactive nature of the AR system may encourage repeated practice, which is important for skill acquisition.

Participants also reported positive perceptions of accessibility, with the statement relating to improving access to sign language learning receiving a mean score of 4.33. In addition, preference for AR-based learning over traditional textbook methods was rated positively ($M = 4.22$), suggesting that participants generally favoured the interactive learning approach.

The questionnaire results are further illustrated in Figure 7, which presents the mean ratings for each statement along with error bars representing the standard error of the mean (SEM). These error bars provide an indication of the variability and reliability of the responses. Most items display relatively small error margins, indicating consistent responses across participants. Slightly larger error bars for items such as precision tracking suggest greater variation in user experiences.

Evaluation Statement	Mean Score (1–5)
The AR interface helped me clearly understand the orientation of the hand signs	4.33
The ability to view the signs from different angles improved my understanding	4.33
The system helped me recognise subtle finger positioning details	4.33
The precision tracker on the app helped improve my sign language practice	4.11
I had to concentrate very hard to understand the signs	2.30
The AR learning method kept me engaged while practising signs	4.33
The application motivated me to continue practising sign language	4.44
The system could make sign language learning more accessible	4.33
I would prefer learning sign language using AR over traditional textbook methods	4.22

Table 1: Summary of questionnaire responses (n = 14)

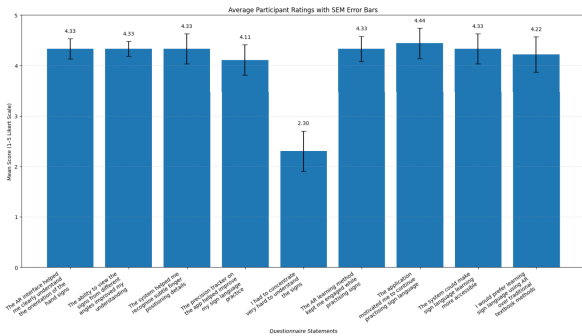


Figure 7: Average Participant Ratings for each Questionnaire Statements

Overall, the results presented in Table 1 and Figure 7 demonstrate a consistent pattern of positive responses across all evaluation measures. The highest scores were associated with motivation, engagement, and gesture visualisation, suggesting that these features were particularly effective in supporting learning. These findings align with prior research indicating that interactive and immersive technologies can enhance learner engagement and improve educational outcomes [6].

5.7 Qualitative Feedback

In addition to the quantitative questionnaire, participants were invited to provide open-ended feedback describing their experience using the system, including what they found useful, challenging, or different compared to their expectations of learning sign language independently.

The qualitative responses were analysed using a thematic analysis approach. An open coding process was first applied, where individual responses were reviewed and labelled with descriptive codes representing key ideas (e.g., “visual clarity”, “feedback usefulness”, “ease of use”). These codes were then grouped into broader themes through an iterative process, allowing recurring patterns in user experience to be identified. This approach enables

a structured interpretation of qualitative data while remaining grounded in participant responses.

A prominent theme was the role of **visual representation** in supporting gesture understanding. Participants frequently emphasised that the system made it easier to interpret hand shapes and movements.

For example, one participant described the system as:

“more visual and interactive, making the signs easier to understand”

Another participant noted:

“it helped me understand the hand movements more clearly”

These responses suggest that the combination of real-time hand tracking and visual overlays allowed users to better perceive spatial aspects of gestures, such as finger positioning and motion paths. This is particularly important in sign language learning, where small differences in orientation or movement can significantly affect meaning. The ability to observe gestures dynamically, rather than relying on static images, appears to have improved users’ conceptual understanding of how signs are formed.

Another key theme was the **usefulness of real-time feedback** in guiding user performance. Participants reported relying on multiple feedback signals provided by the system, including gesture classification outputs and confidence scores.

One participant explained:

“the feedback helped me know if I was doing the sign correctly”

From participant responses, it appears that users primarily interpreted correctness through classification results and confidence values, using these as indicators of whether their gesture matched the expected sign. Some participants also used the visual hand tracking overlays to adjust finger positioning and hand orientation.

This suggests that the system supported an iterative learning process, where users refined their gestures based on immediate feedback. This aligns with prior research demonstrating that timely feedback is critical for effective skill acquisition [10]. However, variability observed in the quantitative results for precision tracking suggests that not all users interpreted or trusted this feedback equally.

Participants also highlighted the **ease of accessing and using the system** as a positive aspect of the design. In particular, the browser-based implementation was perceived as convenient and flexible.

One participant noted:

“it’s easy to use on a laptop or phone”

This suggests that the system’s deployment approach may reduce practical barriers to entry, allowing users to engage with sign language learning without requiring specialised equipment or software installation. While this does not directly measure accessibility in a formal sense, it indicates that the system supports ease of access, which is an important factor in self-directed learning.

Another recurring theme was the **engaging nature of the system**. Participants reported that the interactive interface made the learning process more enjoyable compared to passive methods.

For example:

“it was more engaging than just reading or looking at pictures”

This suggests that the system supports active learning, where users are directly involved in performing gestures rather than passively observing them. Engagement is particularly important in skill-based learning contexts, as it can encourage repeated practice and sustained use over time.

Overall, the qualitative analysis indicates that participants valued the system’s ability to support visual understanding, interactive feedback, and hands-on practice. These themes reflect the extent to which the system achieved its design goals of providing an intuitive and feedback-driven learning experience.

Most Importantly, the findings also highlight how users interpreted and engaged with the system’s feedback mechanisms. While the system appears effective in supporting initial learning and exploration, some variability in responses suggests that further refinement may be needed to improve clarity and consistency of feedback interpretation.

5.8 Discussion

The results of the user evaluation indicate that the proposed AR-based sign language learning system provided a positive and engaging learning experience for beginner users. Both quantitative and qualitative findings suggest that participants were able to better understand and practise gestures using the interactive system. However, beyond demonstrating overall effectiveness, it is important to critically examine how specific design choices within the system contributed to these outcomes.

One of the most notable findings from the study is the high level of user engagement reported by participants, with motivation to continue practising receiving the highest average rating. This can be attributed not only to the use of AR as a medium, but also to the way interaction was structured within the system. Users were able to actively perform gestures and receive immediate responses, creating a continuous feedback loop. This contrasts with traditional learning resources, where interaction is passive and delayed. The inclusion of real-time visual overlays and gesture recognition encouraged users to repeatedly attempt gestures, which is critical for motor skill acquisition. Prior work has shown that interactive and immersive environments can enhance motivation by promoting active participation and immediate reinforcement [29]. However, it is likely that the effectiveness observed here is not solely due to immersion, but also due to the tight coupling between action and feedback implemented in the system.

A key design feature of the system is the real-time feedback mechanism. In this prototype, feedback was delivered through multiple channels: classification labels indicating the recognised gesture, confidence scores representing prediction certainty, and visual hand landmark overlays showing tracked joint positions. Participants responses suggest that this multimodal feedback was useful for understanding gesture accuracy, but the evaluation also highlights limitations in how feedback was interpreted. For example, while confidence scores provide a quantitative indication of correctness, they do not explicitly explain what aspect of the gesture is incorrect. Similarly, skeletal overlays show hand position but require users to interpret discrepancies themselves. This suggests that while immediate feedback was beneficial, it was largely implicit rather than instructional. Future systems could improve this by incorporating more explicit corrective feedback, such as

highlighting incorrect joints, providing directional guidance, or offering step-by-step corrections.

The findings related to gesture clarity and understanding further reflect the impact of specific visual design decisions. The system allowed users to view gestures through tracked hand representations and practise them in real time, which helped participants understand spatial orientation and finger positioning. This supports existing research demonstrating that spatial visualisation tools improve comprehension of complex movements [3]. However, it is important to note that the current implementation relies on 2D screen-based visualisation rather than fully immersive AR. While effective, this may limit depth perception and occlusion handling compared to head-mounted AR systems. Future work could explore true 3D visualisation techniques or multiple viewpoint rendering to further enhance spatial understanding.

The precision tracking component also contributed positively to user experience, but the variability observed in responses suggests that its effectiveness was influenced by technical limitations. The system relies on MediaPipe-based hand tracking, which, while efficient and accessible, can be sensitive to lighting conditions, occlusion, and camera positioning. As a result, inconsistencies in tracking may have affected user trust in the feedback. This highlights an important trade-off in system design: prioritising accessibility and real-time performance over high-precision tracking. Future systems may benefit from combining lightweight tracking with more robust temporal smoothing or model-based correction to improve reliability without sacrificing usability.

Another important consideration is the lack of explicit temporal feedback for dynamic gestures. While Dynamic Time Warping (DTW) was used internally to evaluate gesture sequences, this information was not fully exposed to users in an interpretable way. Participants therefore received limited insight into how their motion compared to the target gesture over time. This suggests an opportunity for future work to visualise temporal alignment or provide feedback on motion trajectories, enabling users to better understand timing and movement patterns, which are critical aspects of sign language.

Participants also perceived the system as easy to access and use across common devices, which reflects the decision to implement the application as a browser-based system. This design choice removes installation barriers and supports cross-platform use, making it suitable for independent learners. However, this accessibility comes with constraints, including limited computational resources and reliance on webcam input. Future work could explore hybrid approaches that maintain accessibility while leveraging more advanced sensing capabilities when available.

Overall, the findings demonstrate that the effectiveness of the system is not solely due to the use of AR, but rather the combination of interactive practice, real-time feedback, and accessible deployment. While the prototype successfully supports engagement and basic gesture learning, the results also highlight areas where more informative feedback, improved tracking robustness, and enhanced temporal guidance could further strengthen the learning experience. These insights provide a clear direction for future development of AR-based sign language tutoring systems.

5.9 Limitations and Lessons Learned

While the evaluation results indicate that the proposed AR-based sign language learning system provides a positive and engaging

user experience, several limitations of both the study design and the prototype system must be acknowledged. These limitations are important not only for interpreting the findings appropriately, but also for identifying key areas for future improvement.

One limitation of the study is the relatively small sample size ($n = 14$), which restricts the generalisability of the findings. While small-scale user studies are common in early-stage Human-Computer Interaction (HCI) research and are effective for identifying usability issues and initial user perceptions [25], they do not provide strong statistical power. In addition, the study primarily relied on self-reported measures of user experience rather than objective measures of learning performance. As a result, while participants perceived improvements in their signing ability, it is not possible to conclusively determine whether their gesture accuracy improved in a measurable way. Future evaluations should incorporate objective performance metrics, such as recognition accuracy, task completion success rates, or pre- and post-test comparisons, to better assess learning outcomes.

A key limitation of the prototype system is its restricted sign vocabulary and simplified representation of sign language. The current implementation focuses primarily on hand gestures, without incorporating non-manual features such as facial expressions, head movement, or body posture, all of which play a crucial role in conveying meaning in sign languages. Furthermore, the system operates on isolated gesture recognition, without consideration of linguistic context. In natural sign language communication, the meaning of a sign is often dependent on surrounding signs, similar to how words in spoken languages rely on sentence context. The absence of contextual understanding limits the system's ability to support more advanced language learning and highlights an important area for future work, such as sequence modelling and contextual recognition.

Technical limitations of the hand tracking and recognition pipeline were also observed. The system relies on MediaPipe-based landmark detection, which, while efficient and accessible, is sensitive to environmental factors such as lighting conditions, camera angle, and occlusion. These limitations can result in inconsistent tracking, which in turn affects gesture recognition accuracy and user confidence in the system. This reflects a broader trade-off in the system design: prioritising real-time performance and accessibility over high-precision sensing. While this decision enables deployment on standard consumer devices, it introduces variability in feedback quality. Future work could explore hybrid approaches, such as incorporating temporal smoothing, multi-frame analysis, or more robust models, to improve reliability without significantly increasing computational requirements [22].

Another important limitation relates to the design of the feedback itself. While the system provides real-time feedback through gesture classification labels, confidence scores, and visual hand tracking overlays, this feedback is largely implicit. Users must interpret the information themselves to determine what aspects of their gesture are incorrect. The evaluation results suggest that while this form of feedback is useful, it may not be sufficient for all learners, particularly beginners who may struggle to diagnose errors. This highlights a key lesson learned: effective feedback in educational systems should not only be immediate, but also informative and actionable. Future systems could improve this by providing explicit corrective guidance, such as highlighting incorrect finger positions, suggesting adjustments, or breaking gestures down into smaller steps.

The development process also highlighted the importance of balancing technical capability with usability. While more complex recognition models could improve gesture accuracy, they may also introduce latency or increase system requirements, which would negatively impact user experience. The decision to implement a lightweight, browser-based system enabled ease of access and cross-platform compatibility, but also constrained the complexity of the models that could be deployed. This trade-off emphasises the importance of designing systems that are not only technically effective but also practical and accessible for real-world use.

Finally, an important lesson learned relates to the role of interaction design in supporting learning. The system's effectiveness appears to stem not only from the use of AR, but from the integration of interactive practice, real-time feedback, and visualisation. Participants were able to iteratively perform gestures, observe the system's response, and adjust their actions accordingly. This iterative feedback loop is critical for motor skill learning and suggests that future systems should prioritise continuous interaction and guided practice rather than passive content delivery.

Overall, while the prototype demonstrates the feasibility and potential benefits of AR-assisted sign language learning, the limitations identified in this study highlight important challenges in recognition accuracy, feedback design, and linguistic completeness. Addressing these challenges will be essential for developing more robust, scalable, and pedagogically effective sign language learning systems in future work.

6 Conclusions

This project explored the design and evaluation of an augmented reality (AR)-based system to support the learning of British Sign Language (BSL). The proposed approach combined real-time hand tracking, lightweight machine learning, and interactive visual feedback within a browser-based environment to enable accessible, self-guided practice.

The evaluation findings indicate that the system was effective in supporting beginner learners, particularly in improving understanding of hand orientation, finger positioning, and overall gesture clarity. Participants reported high levels of engagement and motivation, suggesting that the interactive and feedback-driven design encouraged repeated practice. These findings reinforce the importance of combining real-time feedback with active user interaction when designing systems for motor skill learning [21, 29].

A key contribution of this work is the demonstration that sign language learning support can be delivered using lightweight, widely accessible technologies. By operating entirely within a web browser and leveraging efficient computer vision techniques, the system reduces barriers to entry and supports use across a range of devices. In addition, the integration of both static gesture recognition and dynamic gesture evaluation using Dynamic Time Warping provides a foundation for supporting more complex aspects of sign language beyond static pose matching.

Overall, this research demonstrates the feasibility of AR-assisted sign language learning and highlights the potential of interactive, real-time systems to improve engagement and perceived learning effectiveness. The findings contribute to ongoing work in educational technology by illustrating how accessible, vision-based tools can support the acquisition of visually complex motor skills such as sign language.

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